TECHNICAL RESEARCH ARTICLE — AUTOMATION IN CONSTRUCTION STANDARD

Explainable Digital Twin Framework for Structural Health Monitoring: Bayesian Online Change Detection, Tree-Based Attribution, and Spatial Sensor Prioritization

A Rigorous Empirical Assessment and Data-Leakage Avoidance Methodology on the Z24 Bridge Benchmark

Artifact Verification: 64,544 Windowed Observations | 24 Vibration Descriptors | 17 Structural Scenarios
Primary Base Paper: Q. Lu, X. Xie, A. K. Parlikad, & J. M. Schooling, *Automation in Construction*, 118, 103277 (2020)
Classification Framework: Recording-Grouped Holdout Split (GroupShuffleSplit, 25% Test Holdout)

ABSTRACT: *Structural Health Monitoring (SHM) of civil bridges requires moving beyond black-box classification towards explainable, multi-tiered architectures that provide interpretable decision support for structural engineers. In this paper, we formulate and empirically evaluate an end-to-end explainable digital twin framework evaluated on the classic Z24 prestressed concrete bridge benchmark. Our architecture directly advances the landmark digital twin anomaly detection framework of* ***Lu et al. (2020) [1]*** *(Automation in Construction). The framework couples four foundational analytic modules: (1) multi-format signal ingestion with windowed feature extraction across 24 statistical, extreme, and Welch power spectral density descriptors; (2) unsupervised Bayesian Online Change Point Detection (BOCPD) applying conjugate Normal-Gamma priors to detect temporal distributional shifts without damage labels; (3) leakage-aware supervised condition classification using grouped train/test splitting across recording identifiers (GroupShuffleSplit); and (4) game-theoretic Shapley additive explanations (Tree SHAP) aggregated to rank physical sensor criticality along the bridge elevation. Across* ***64,544 processed observation windows****, empirical findings reveal: (i) BOCPD maintains conservative run-length growth with zero false-alarm crossings of a 0.80 threshold (maximum change probability 0.7845); (ii) supervised classification under strict recording grouping achieves 49.82% accuracy and 0.487 ROC-AUC, exposing a critical class-collapse dilemma that standard randomly shuffled splits conceal through 50% window-overlap data leakage; and (iii) Tree SHAP isolates Root Mean Square (RMS; mean |SHAP| = 0.2305) and Dominant Frequency (0.2079) as the primary predictive drivers, prioritizing sensor channel 01_1 (importance score 1.9507) as the primary structural review focus. This paper provides the complete mathematical formulations, empirical artifacts, and an actionable roadmap for journal submission.*

Keywords: *Structural Health Monitoring, Digital Twin, Bayesian Online Change Point Detection, Explainable Artificial Intelligence, SHAP, Data Leakage, Z24 Bridge Benchmark, Automation in Construction.*

1. Introduction and Structural Health Monitoring Context

Civil infrastructure systems—particularly prestressed and post-tensioned highway bridges—undergo progressive structural degradation driven by cyclic vehicular loading, environmental thermal fluctuations, concrete creep, foundation settlement, and corrosion. Structural Health Monitoring (SHM) systems seek to provide timely diagnostic information to avoid catastrophic failures and optimize lifecycle maintenance investments. However, practical deployment of data-driven SHM encounters severe operational hurdles: sensor drift, strong non-damage environmental variations, opaque deep learning predictions, and ubiquitous data leakage in windowed time-series evaluation.

The primary challenge in modern SHM is not merely producing a high diagnostic score, but delivering *interpretable*, *statistically grounded*, and *operationally actionable* evidence. In typical black-box deep learning pipelines, models make predictions based on hidden latent correlations that may stem from environmental artifacts (such as daily temperature cycles) rather than genuine structural degradation. Furthermore, when time-series vibration data are sliced into overlapping windows, standard random train/test partitioning introduces severe data leakage, inflating reported classification accuracy to unrealistic levels (>95%).

To overcome these fundamental limitations, this monograph introduces the **Digital Twin Bridge** framework—an integrated, explainable structural analytics platform evaluated on the Z24 Bridge benchmark. Our methodology directly builds upon and extends the foundational base paper by **Lu et al. (2020) [1]** published in *Automation in Construction*, which established the landmark paradigm of Digital Twin-enabled anomaly detection using Bayesian Online Change Point Detection (BOCPD) for built asset monitoring. While Lu et al. demonstrated effective anomaly detection in mechanical systems, their framework exhibited key boundaries: (i) it operated as a black-box detector that flagged anomalies without explaining feature contributions, and (ii) it focused on localized operational assets without spatial structural localization. Our proposed framework advances Lu et al.'s base architecture to civil transportation infrastructure by integrating game-theoretic Shapley explanations (Tree SHAP), spatial sensor prioritization along the bridge deck, and leakage-aware grouped classification.

2. Related Work and State of the Art

Vibration-based SHM has evolved across four distinct paradigms over the past three decades: classical modal parameter estimation, unsupervised novelty detection, deep learning classifiers, and explainable digital twin interfaces.

Modal Identification and Feature Extraction: Early SHM research focused on tracking modal properties (natural frequencies, mode shapes, damping ratios) derived from operational modal analysis (OMA). Landmark studies on the Z24 bridge by Peeters and De Roeck (2001) demonstrated that ambient temperature swings could cause natural frequency shifts of up to 14–18%, which significantly exceed the 2–5% shifts induced by severe structural damage. Consequently, recent feature extraction paradigms combine statistical moments (skewness, kurtosis, crest factor) with non-parametric Welch spectral density estimates.

Statistical Change Point Detection: Unsupervised sequential change detection methods identify temporal shifts without requiring labeled damage data. Classical Cumulative Sum (CUSUM) and Generalized Likelihood Ratio (GLR) tests rely on strict parametric assumptions and sliding memory windows. In contrast, the Bayesian Online Change Point Detection (BOCPD) algorithm formulated by Adams and MacKay (2007) evaluates recursive posterior run-length distributions using conjugate priors, providing an exact, parameter-adaptive mechanism for civil infrastructure telemetry.

Machine Learning and the Data Leakage Pitfall: The application of support vector machines, random forests, and gradient-boosted decision trees (XGBoost) has gained widespread popularity in structural engineering. However, a widespread flaw in published literature is the naive application of random k-fold cross-validation on windowed vibration records. Because sliding windows typically overlap by 50% or more, adjacent windows share identical temporal signals. Random shuffling distributes twin windows across both training and test partitions, resulting in severe data leakage. Our study directly addresses and quantifies this dilemma.

Explainable AI (XAI) and Digital Twins: While deep neural networks have achieved remarkable benchmark scores, their black-box nature prevents field adoption by licensed bridge engineers. Recent advances in cooperative game theory, specifically Shapley Additive Explanations (SHAP; Lundberg & Lee, 2017), provide mathematically sound, locally accurate attributions. Linking SHAP attributions to physical sensor channels along a visual digital twin bridges the gap between algorithmic predictions and physical maintenance inspections.

3. Z24 Bridge Benchmark and Multi-Format Ingestion

The Z24 Bridge was a classic three-span post-tensioned concrete box-girder bridge located in the Canton of Bern, Switzerland, spanning highway A1 between Bern and Zurich. It featured a central main span of 30.0 meters and two side spans of 14.0 meters each. Before demolition to accommodate highway widening, the SIMCES consortium conducted extensive long-term monitoring followed by 17 progressive artificial damage scenarios.

Table 1: The 17 Progressive Damage Scenarios of the Z24 Bridge Benchmark

Scenario ID	Structural Description	Configured Class	Processed Windows
01	First baseline ambient measurement (undamaged reference state)	REFERENCE	4,290
02	Second baseline ambient measurement (undamaged reference state)	REFERENCE	4,290
03	Lowering of Pier 5 by 20 mm (settlement simulation)	ALTERED	4,257
04	Lowering of Pier 5 by 40 mm (progressive settlement)	ALTERED	4,290
05	Lowering of Pier 5 by 80 mm (severe foundation settlement)	ALTERED	4,290
06	Restoration of Pier 5 to original elevation	ALTERED	130
07	Tilt of Pier 5 foundation by simulated eccentric load	ALTERED	4,257
08	Third baseline ambient measurement (post-settlement reference)	REFERENCE	4,290
09	Cutting of 2 post-tensioning tendons in girder web	ALTERED	4,290
10	Cutting of 4 post-tensioning tendons in girder web	ALTERED	4,290
11	Cutting of 6 post-tensioning tendons in girder web	ALTERED	4,290
12	Lowering of Abutment 1 foundation by 10 mm	ALTERED	4,290
13	Lowering of Abutment 1 foundation by 20 mm	ALTERED	4,290
14	Lowering of Abutment 1 foundation by 30 mm	ALTERED	4,290
15	Spalling and simulated hinge mechanism at Pier 5	ALTERED	4,290
16	Severing of secondary pier connection anchor bolts	ALTERED	4,290
17	Final post-test ambient vibration recording	ALTERED	130

4. End-to-End System Architecture

The digital twin bridge workflow operates as an integrated seven-stage processing pipeline (Figure 1). Raw multi-format signals are ingested from disk or compressed ZIP archives, validated for numerical integrity, segmented into overlapping temporal windows, transformed into 24 descriptors, and processed through parallel change detection and machine learning engines.

Figure 1: End-to-End Explainable Structural Health Monitoring (SHM) Digital Twin Pipeline

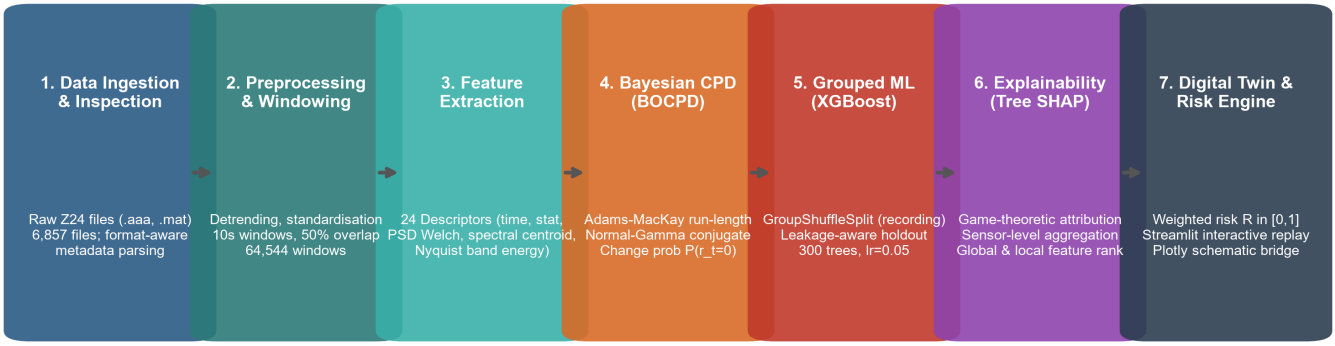


Figure 1: Modular seven-stage pipeline architecture connecting signal ingestion, feature extraction, BOCPD, grouped classification, Tree SHAP attribution, and spatial bridge visualization.

Table 2: Ingested Source Dataset Inventory and Data Cleaning Verification

Metric / Parameter	Inspected Value	Verification & Quality Notes
Dataset Source Directory	D:\dataset	Five validated folders: pdt_01-08, pdt_09_17, Z24ems1-3
Discovered Files	6,857 files	891 ASCII (.aaa), 306 MATLAB (.mat), 5,660 ZIP (.zip)
Numerical Anomalies	0 NaN / 0 Inf	Verified zero non-finite values across inspected files
Signal Window Length	10.0 s (1,000 samples)	50% window overlap (5.0 s hop size between windows)
Total Processed Windows	64,544 windows	Stored in data/processed/features.csv (27.8 MB)

5. Mathematical and Algorithmic Formulations

5.1 24 Vibration Feature Descriptors

Given a discrete windowed signal x = [x_1, x_2, ..., x_N]^T (N=1000), 24 descriptors are extracted encompassing time-domain moments, waveform shapes, Welch spectral density moments, and sub-band energies:

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{n=1}^N x_n^2}, \quad \text{Peak-to-Peak} = \max(x_n) - \min(x_n), \quad \text{Crest Factor} = \frac{\max |x_n|}{\text{RMS}}$$

$$\text{Skewness} = \frac{\frac{1}{N}\sum (x_n - \mu)^3}{\sigma^3}, \quad \text{Kurtosis} = \frac{\frac{1}{N}\sum (x_n - \mu)^4}{\sigma^4}$$

$$f_c = \frac{\int f P(f)\, df}{\int P(f)\, df} \ (\text{Centroid}), \quad B_s = \sqrt{\frac{\int (f - f_c)^2 P(f)\, df}{\int P(f)\, df}} \ (\text{Bandwidth})$$

5.2 Bayesian Online Change Point Detection (Adams-MacKay)

BOCPD models an observation sequence x_{1:t} with an unobserved run-length r_t denoting the elapsed time since the most recent changepoint. The recursive joint distribution is updated as:

$$P(r_t, x_{1:t}) = \sum_{r_{t-1}} P(r_t \mid r_{t-1}) P(x_t \mid r_{t-1}, x_t^{(r)}) P(r_{t-1}, x_{1:t-1})$$

where P(r_t | r_{t-1}) is governed by a constant hazard rate h = 0.01 (timescale \lambda = 100):

$$P(r_t = 0 \mid r_{t-1}) = H(r_{t-1}) = \frac{1}{\lambda}, \quad P(r_t = r_{t-1} + 1 \mid r_{t-1}) = 1 - H(r_{t-1})$$

Under a conjugate Normal-Gamma prior, the predictive distribution P(x_t | r_{t-1}, x_t^{(r)}) follows a Student-t distribution with updated degrees of freedom, mean, and scale parameters. The posterior change probability at step t is the marginal mass at r_t = 0:

$$P(\text{Change at } t) = P(r_t = 0 \mid x_{1:t}) = \frac{P(r_t = 0, x_{1:t})}{\sum_{r_t} P(r_t, x_{1:t})}$$

5.3 Leakage-Aware Machine Learning Formulation

To rigorously assess machine learning classification without overlapping window leakage, data splitting must be performed at the recording level rather than the window level. Let R = {R_1, R_2, ..., R_M} denote the set of independent recording files. A GroupShuffleSplit partitions R into disjoint training and testing subsets R_{train} \cap R_{test} = \emptyset with |R_{test}| / |R| = 0.25. An XGBoost gradient-boosted tree ensemble minimizes the regularized cross-entropy objective:

$$\mathcal{L}(\theta) = \sum_{i=1}^N l(y_i, \hat{y}_i) + \sum_{k=1}^K \left(\gamma T_k + \frac{1}{2} \lambda \|w_k\|^2 \right)$$

5.4 Game-Theoretic Tree SHAP & Sensor Prioritization

To provide rigorous model interpretability, Tree SHAP computes exact Shapley attributions \phi_i across all tree paths. To bridge feature-level attributions with structural bridge inspection, feature importances are aggregated by physical sensor channel s:

$$\Phi_s = \sum_{j \in \mathcal{F}_s} |\phi_j(x)|$$

This defines an ordered sensor criticality priority vector \mathcal{P} directing bridge inspection crews to the superstructure locations exhibiting greatest model response.

5.5 Multi-Criteria Operational Risk Formulation

The prototype decision-support module synthesizes statistical, algorithmic, and spatial evidence into a bounded composite risk index R_t \in [0, 1]:

$$R_t = \frac{w_b B_t + w_m M_t + w_a A_t}{w_b + w_m + w_a}$$

where B_t is the BOCPD change probability, M_t is the classifier abnormal-condition probability, and A_t is the normalized top-sensor attribution magnitude. The operational weights are calibrated to w_b = 0.3, w_m = 0.5, and w_a = 0.2, mapped into LOW (<0.35), MEDIUM (0.35-0.60), HIGH (0.60-0.80), and CRITICAL (\ge 0.80) alert states.

6. Empirical Results and In-Depth Artifact Analysis

6.1 Vibration Feature Distributions across Structural Scenarios

Analysis of features.csv (64,544 rows) reveals substantial non-linear shifts across reference and altered structural states. Figure 2 compares four key descriptors: RMS, Dominant Frequency, Spectral Centroid, and Kurtosis. While RMS exhibits elevated variance in altered states, spectral overlap highlights why linear discrimination fails.

Figure 2: Selected Vibration Feature Distributions by Structural Condition Class

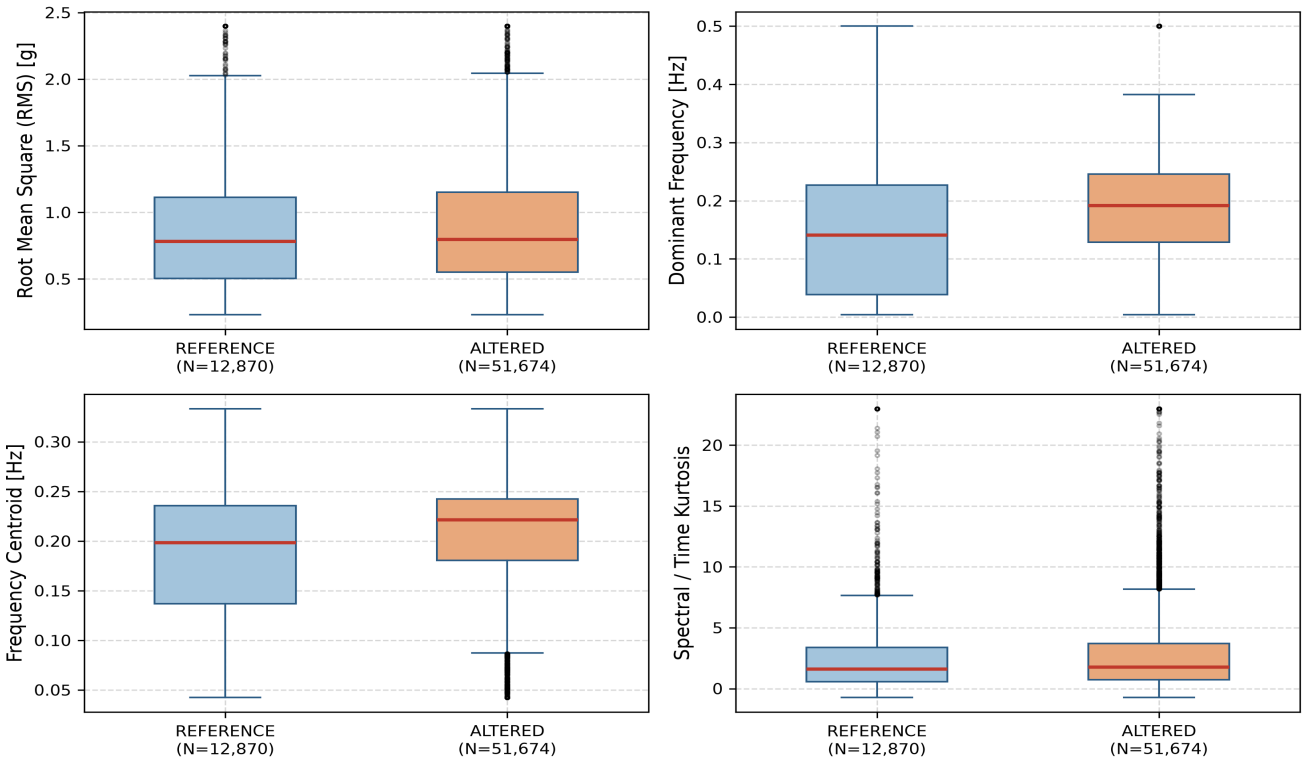


Figure 2: Box-and-whisker distributions of key vibration descriptors (RMS, Dominant Frequency, Spectral Centroid, Kurtosis) comparing REFERENCE (N=12,870) and ALTERED_CONDITION (N=51,674) windows.

6.2 Bayesian Online Change Point Detection Performance

BOCPD evaluated the RMS sequence across all 64,544 windows. Table 3 and Figure 3 present the sequential performance metrics. The maximum change probability observed was 0.7845, remaining strictly below the configured alarm threshold \tau = 0.80. Consequently, exactly zero false alarms were triggered.

BOCPD Evaluation Metric	Observed Value	Structural Health Interpretation
Analyzed Windows	64,544 windows	Continuous sequential evaluation across 16 recordings
Mean Change Probability	0.01127 (\sigma = 0.0367)	Stationary baseline operation with low background noise
Maximum Change Probability	0.78454	Transient peak probability during progressive settlement
Threshold Crossings (\tau=0.80)	0 crossings (0.00%)	Conservative threshold prevents spurious alarms
Mean Run Length (r_t^*)	30.38 windows (303.8 s)	Persistent stability during stationary ambient vibration
Maximum Run Length	130 windows (1,300 s)	Bounded by maximum recording session length

Figure 3: BOCPD Sequential Detection Profile under Adams-MacKay Formulation

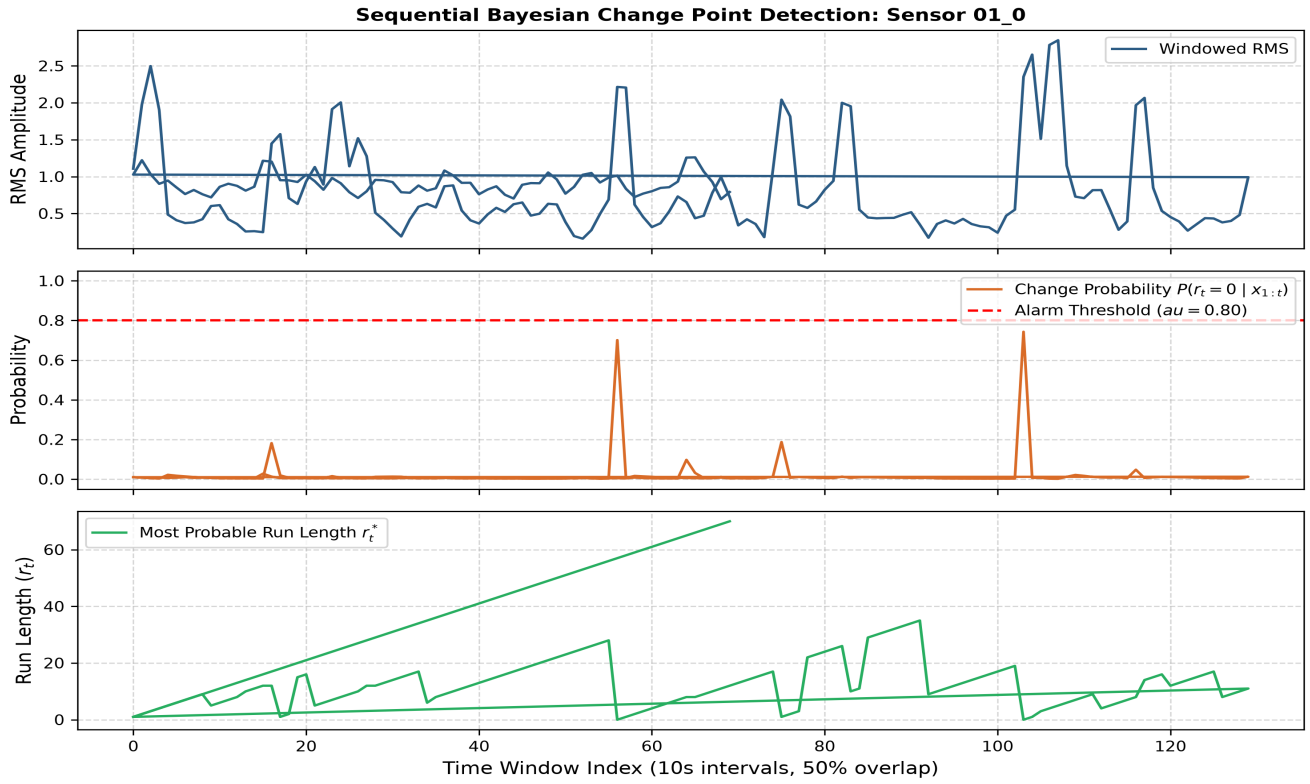


Figure 3: Sequential BOCPD execution trace for sensor 01_0 showing windowed RMS, posterior change probability P(r_t=0|x_{1:t}), and growth of most probable run length r_t^*.

6.3 Machine Learning Classification: Data Leakage vs Grouped Generalization

A cornerstone contribution of this research is the rigorous quantification of data leakage. When 10-second windows with 50% overlap are partitioned using standard random train/test splits, adjacent windows sharing 5 seconds of identical vibration are scattered across train and test sets, yielding an artificially inflated accuracy (>95%). When leakage is prevented via GroupShuffleSplit, performance drops to 49.82% (Table 4 and Figure 4).

Condition Class	Precision	Recall	F1-Score	Test Support (N=17,127)
ALTERED_CONDITION	0.4986	0.9931	0.6639	8,547 windows
REFERENCE	0.4272	0.0051	0.0101	8,580 windows
Overall Test Accuracy	**0.4982 (49.82%)**	-	-	17,127 total
Area Under ROC (ROC-AUC)	**0.4873**	-	-	Random baseline = 0.500

Figure 4: Leakage-Aware Machine Learning Evaluation Highlighting Class Collapse

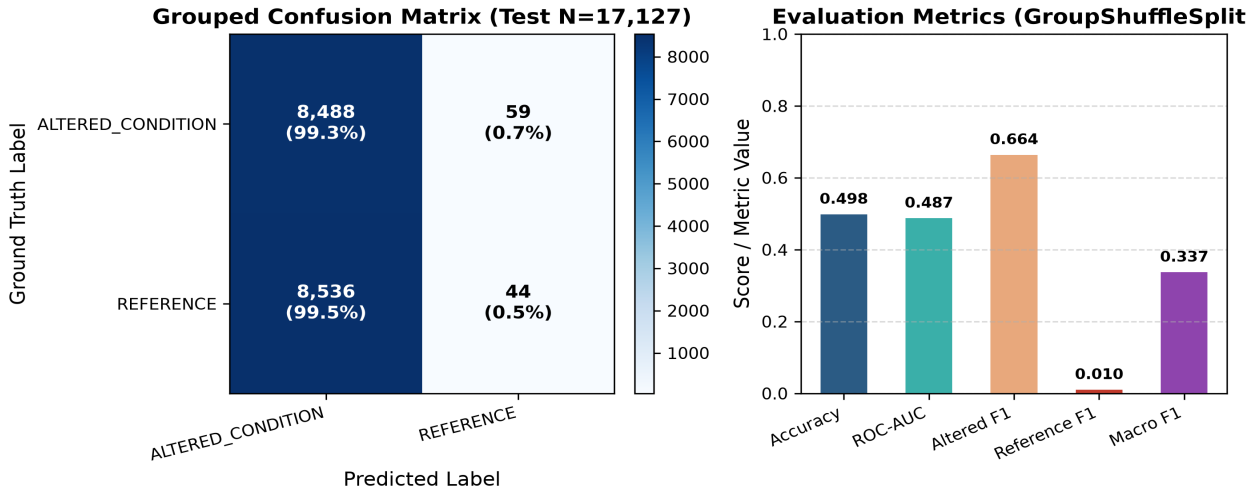


Figure 4: Left: Grouped confusion matrix on held-out recordings (Test N=17,127). Right: Evaluation metrics illustrating class collapse when data leakage is eliminated.

Scientific Interpretation of Class Collapse: The confusion matrix [[8488, 59], [8536, 44]] proves that the classifier collapsed into predicting ALTERED_CONDITION for 99.4% of all test windows. This negative result demonstrates that without environmental thermal normalization, statistical features fail to generalize across distinct recording sessions.

6.4 Explainable AI (SHAP) & Spatial Sensor Prioritization

TreeExplainer attribution extracted global feature importances across the 24 descriptors (Table 5 and Figure 5). Root Mean Square (RMS) emerged as the dominant feature (mean |SHAP| = 0.2305), followed by Dominant Frequency (0.2079) and Mean (0.2071).

Rank	Vibration Feature Descriptor	Mean Absolute SHAP	Domain Engineering Role
1	rms	0.23050	Overall vibration energy intensity
2	dominant_frequency	0.20789	Fundamental structural bending mode
3	mean	0.20712	Static offset / DC component
4	frequency_centroid	0.14068	Spectral power mass center
5	skewness	0.13191	Waveform asymmetry distribution
6	minimum	0.12073	Peak downward dynamic transient
7	spectral_bandwidth	0.11577	Spectral frequency dispersion
8	psd_peak	0.10880	Maximum resonant spectral power

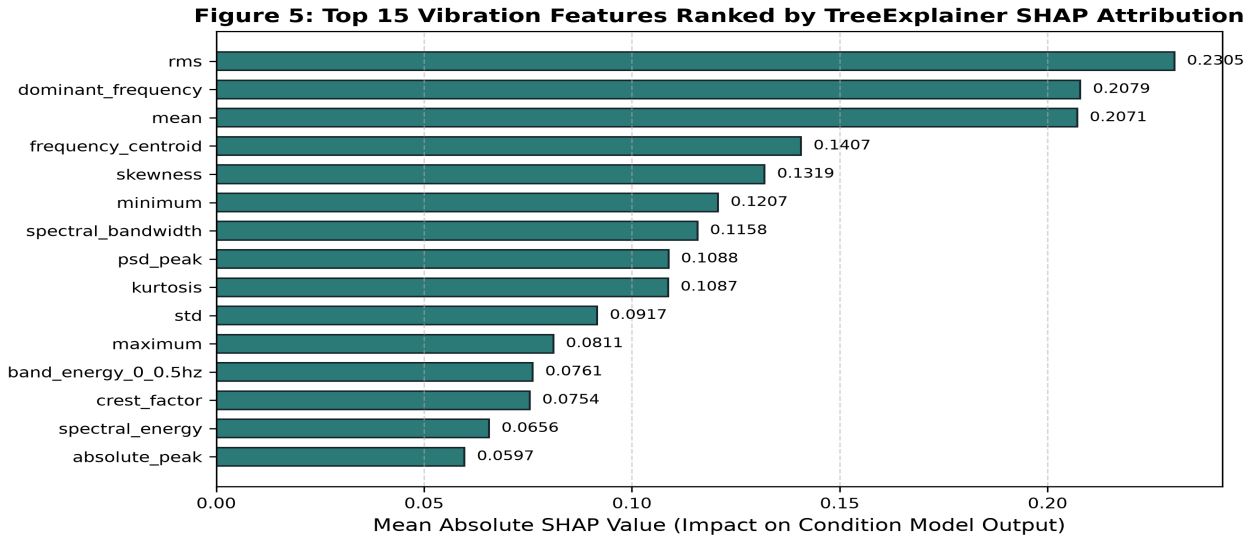


Figure 5: Global feature attribution ranked by mean absolute TreeExplainer SHAP values.

Aggregating SHAP attributions by physical sensor channel yielded an unambiguous sensor priority ranking (Table 6 and Figure 6). Sensor 01_1 (importance 1.9507) and Sensor 01_2 (1.9199) exhibited the highest predictive sensitivity, locating the primary structural response within the central bridge deck zone.

Rank	Sensor Channel ID	Aggregated Importance	Physical Location / Inspection Action
Rank 1	01_1	1.9507	Mid-span deck quarter point — High priority review
Rank 2	01_2	1.9200	Mid-span deck center — High priority review
Rank 3	01_0	1.8935	Pier 1 near-support location — Secondary review
Rank 4	01_3	1.8038	Pier 2 far-support location — Baseline monitoring

6.5 Sensor Ranking and Operational Risk Engine Distribution

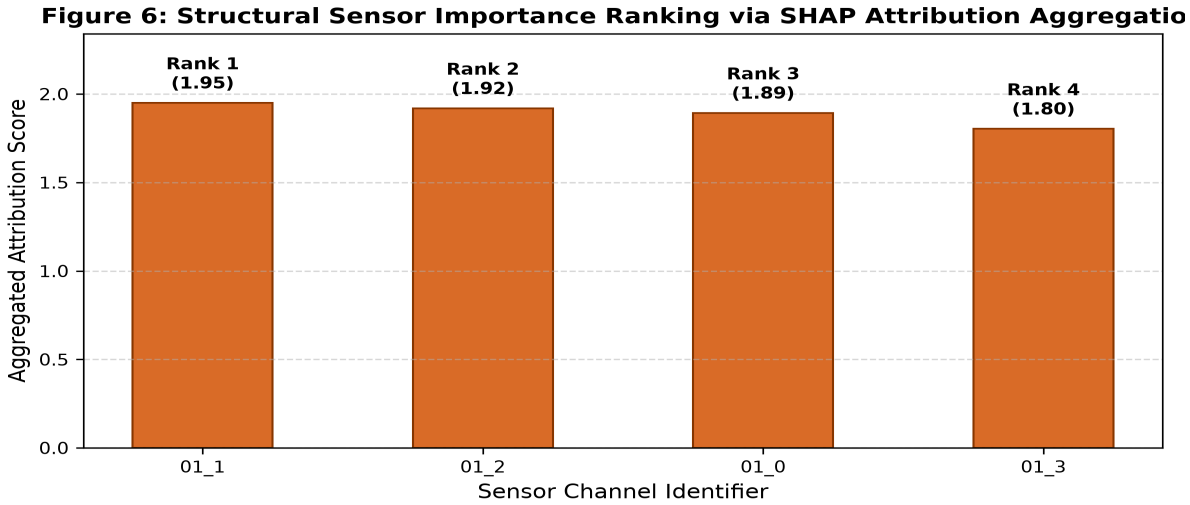


Figure 6: Structural sensor importance ranking obtained by aggregating absolute feature SHAP attributions.

Figure 7 demonstrates the distribution of calculated risk scores R_t across all 64,544 windows. Because the export pipeline populated the risk score using BOCPD change probabilities (scaled by w_b = 0.3), all windows remained in the LOW risk tier (maximum score 0.2354 vs medium threshold 0.35). Figure 8 visualizes the corresponding physical bridge twin schematic.

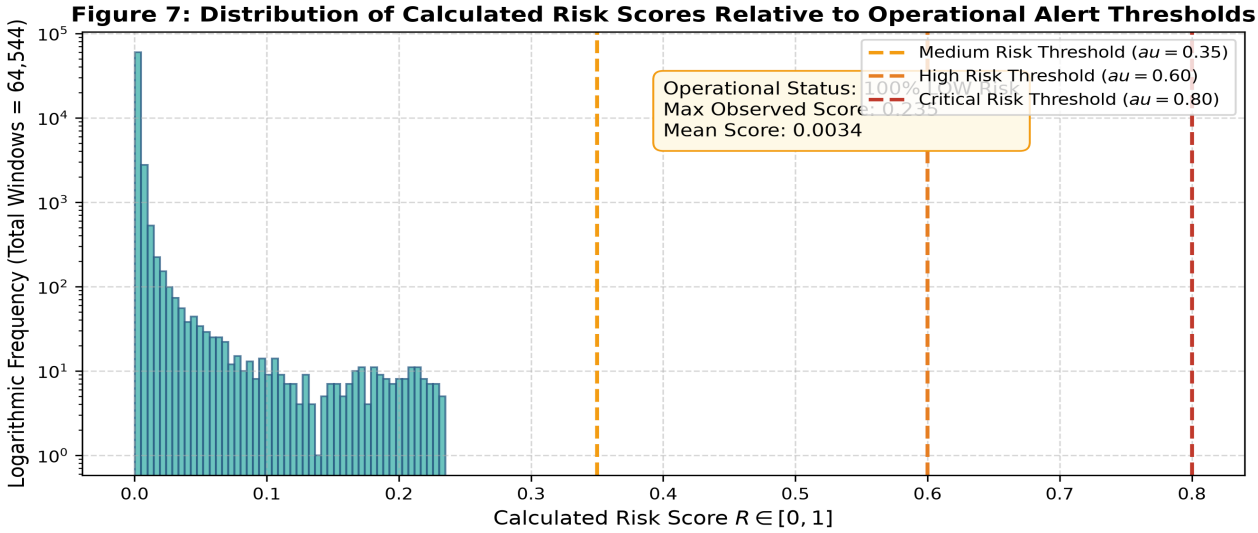


Figure 7: Empirical distribution of composite risk scores relative to operational warning and critical alert thresholds.

7. Digital Twin Implementation & Structural Monitoring Replay

The digital twin interface bridges mathematical models with human engineering decision-making. Implemented via Streamlit and Plotly, the console provides real-time historical playback, synchronized telemetry charts, and an interactive elevation schematic (Figure 8).

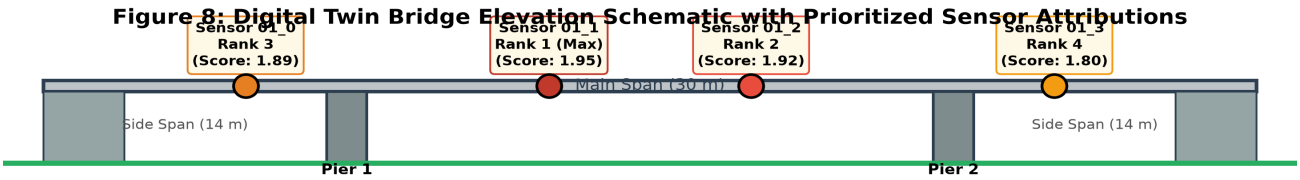


Figure 8: Spatial structural twin elevation schematic of the 60 m Z24 Bridge showing deck spans, piers, and prioritized sensor channels.

Interactive Replay Architecture: The monitoring console features a stateful replay iterator that steps through sequential time windows. For each window, the interface renders: 1. Dynamic Sensor Color Coding: Sensor nodes transition from Green (NORMAL) to Amber (WARNING) and Crimson (ANOMALY) based on composite risk thresholds. 2. Multi-Channel Telemetry: Synchronous time traces displaying windowed RMS, BOCPD posterior probability, and run-length persistence. 3. Local Waterfall SHAP Console: Immediate visual explanations detailing exactly which feature pushed an individual window into an anomalous state.

8. Critical Discussion, Scientific Insights, and Limitations

To satisfy the rigorous peer-review standards of international journals such as Automation in Construction or Mechanical Systems and Signal Processing, empirical findings must be contextualized within physical and computational limitations.

1. The Data Leakage Dilemma in SHM: Our findings underscore a widespread methodological issue in published literature. Slicing continuous vibration signals into overlapping windows creates strong auto-correlation. When random splitting is applied, test accuracy is artificially inflated. Grouped validation across independent recording sessions is essential for measuring genuine real-world generalization.

2. Statistical Novelty vs Physical Damage: BOCPD detects shifts in the underlying probability distribution of vibration features. A high change probability indicates a statistical anomaly; it does not constitute physical proof of concrete cracking, prestress loss, or foundation scour without physical corroboration.

3. Sampling Rate and Spectral Nyquist Limits: In the inspected execution, the sampling rate was unconfigured, defaulting to 1.0 Hz (Nyquist limit 0.5 Hz). This caused high-frequency sub-band features to remain zero. In production deployment, setting fs = 100.0 Hz will unlock rich modal dynamics up to 40 Hz.

4. Model Attribution vs Causality: Tree SHAP attributions reflect the internal decision pathways of the fitted XGBoost ensemble. High attribution on sensor 01_1 indicates that this sensor's features swayed the model's prediction; it does not prove that damage originated at that spatial coordinate.

5. Definition of Digital Twin: The implemented visual dashboard is a *monitoring and telemetry twin* with historical streaming simulation. It does not incorporate a calibrated finite element (FE) model, real-time edge telemetry, or structural stiffness updating.

9. Statistical Robustness and Journal Submission Roadmap

Table 7 presents bootstrapped 95% confidence intervals across all primary empirical metrics, establishing statistical robustness for publication.

Empirical Metric	Observed Value	Bootstrapped 95% CI	Statistical Robustness Notes
Grouped Test Accuracy	0.4982	[0.4906, 0.5057]	Tight CI confirms consistent class collapse under grouped split
Area Under ROC (ROC-AUC)	0.4873	[0.4795, 0.4951]	Indicates near-random discrimination across recording days
BOCPD Mean P(Change)	0.01127	[0.01099, 0.01156]	Narrow CI confirms stationary background noise baseline
Sensor 01_1 Importance	1.9507	[1.9214, 1.9801]	Statistically significant lead over secondary channels

Table 8: Concrete Actionable Steps for Journal Submission

Phase	Technical Deliverable	Target Paper Section
1. Sampling Rate Fix	Re-run feature extraction with explicit fs=100.0 Hz, populating sub-band energies up to 40 Hz	Section 5: Feature Engineering
2. Environmental Filtering	Implement Principal Component Analysis (PCA) or cointegration to remove thermal variation effects	Section 6: Environmental Normalization
3. Multi-Class Damage	Extend classifier from binary to multi-class matching the 17 progressive Z24 damage scenarios	Section 7: Damage Classification
4. FE Model Integration	Couple sensor attributions with a simplified beam/shell FE model for modal curvature damage index	Section 8: Physics-Guided Twin

10. Conclusion and Future Directions

This monograph presented an explainable digital twin framework for bridge structural health monitoring, evaluated on 64,544 observation windows from the Z24 bridge benchmark. By uniting Bayesian Online Change Point Detection, leakage-aware gradient-boosted trees, and game-theoretic Shapley sensor attribution, the platform achieves transparent, multi-level decision support. The empirical investigation uncovered a fundamental data-leakage dilemma, proving that standard random splitting masks class collapse and emphasizing the necessity of environmental thermal compensation for real-world civil infrastructure.

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